

(c) Electromagnetism	
Students should:	
6.8	know that an electric current in a conductor produces a magnetic field around it
6.9P describe the construction of electromagnets	
6.10P draw magnetic field patterns for a straight wire, a flat circular coil and a solenoid when each is carrying a current	
6.11P know that there is a force on a charged particle when it moves in a magnetic field as long as its motion is not parallel to the field	
6.12	understand why a force is exerted on a current-carrying wire in a magnetic field and how this effect is applied in simple d.c. electric motors and loudspeakers
6.13	use the left-hand rule to predict the direction of the resulting force when a wire carries a current perpendicular to a magnetic field
6.14	describe how the force on a current-carrying conductor in a magnetic field changes with the magnitude and direction of the field and current